

Shashwat Shrivastava

Resume

EPFL

+41 762637453

✉ shashwat.shrivastava@epfl.ch

Interests

My work focuses on FPGA compilation and architecture–compiler co-design. I develop compilation techniques that adapt to emerging application requirements and evolving FPGA architectures, enabling FPGA-based solutions with higher performance, lower area, and improved energy efficiency. I am interested in reconfigurable computing, domain-specific architectures, and efficient compilation tools for emerging applications.

Education

- 2021-2026 **Ph.D. in Computer and Communication Sciences**, EPFL, Lausanne
- 2016-2021 **B.Tech. and M.S. by research in Electronics and Communication**, IIT-Hyderabad
- Accelerating stereo vision, local Laplacian filters, and CNNs on FPGAs

Work Experience

- Jun - Dec 2023 **Research Intern**, AMD, Longmont, Colorado, USA
Accelerating FPGA routing for commercial architectures
- Jan - July 2021 **Research Intern**, Intel Labs, Bangalore, India
Hardware-software co-design for 3D image segmentation
- May - August 2019 **Student Developer**, Google Summer of Code Student, Hyderabad, India
Development of dynamic and interactive robotic simulator
- May - July 2018 **Research Intern**, Bluespec Inc., Mysore, India
Hardware design space exploration for dense matrix multiplier

Publications

- FCCM2026 **PathSteiner: Improving PathFinder with Quasi-Optimal Steiner-Tree Initialization**
Shashwat Shrivastava, Luka Kurešević, Alex Poupakis, Chirag Ravishankar, Dinesh Gaitonde, Stefan Nikolić, Mirjana Stojilović
- FCCM2025 **Guaranteed Yet Hard to Find: Uncovering FPGA Routing Convergence Paradox**
Best Paper
Shashwat Shrivastava, Stefan Nikolić, Sun Tanaka, Chirag Ravishankar, Dinesh Gaitonde, Mirjana Stojilović

- ICCAD2023 **IIBLAST: Speeding Up Commercial FPGA Routing by Decoupling and Mitigating the Intra-CLB Bottleneck**
Shashwat Shrivastava, Stefan Nikolić, Chirag Ravishankar, Dinesh Gaitonde, Mirjana Stojilović
- HaSS **Instruction-Level Leakages of Soft-Core CPUs on Shared FPGAs**
 Ognjen Glamočanin, **Shashwat Shrivastava**, Jinwei Yao, Nour Ardo, Mathias Payer and Mirjana Stojilović
- ACM **An FPGA Overlay for CNN Inference with Fine-grained Flexible Parallelism**
- TACO2022 Ziaul Choudhury, **Shashwat Shrivastava**, Lavanya Ramapantulu, Suresh Purini
- FPL2021 **FPGA Accelerator for Stereo Vision using Semi-Global Matching through Dependency Relaxation**
Shashwat Shrivastava, Ziaul Choudhury, Shashwat Khandelwal, and Suresh Purini
- FPL2021 **Accelerating Local Laplacian Filters on FPGAs**
 Shashwat Khandelwal, Ziaul Choudhury, **Shashwat Shrivastava**, and Suresh Purini

Teaching Experiences

EPFL

- Fundamentals of Digital Systems, instructor: Dr. Mirjana Stojilović
- Information, Computation, Communication, instructor: Dr. Mirjana Stojilović
- Computer Architecture, instructor: Dr. Mirjana Stojilović

IIIT-H

- Intro to Psychology, instructor: Prof. Priyanka Shrivastava, Fall-2020
- Intro to Processor Architecture, instructor: Prof. Suresh Purini, Spring-2020
- Technology Product Entrepreneurship, instructor: Ramesh Loganathan, Fall-2019
- Intro to Internet of Things, instructor: Dr. Lavanya Ramapantulu, Spring-2019
- Embedded Hardware Design, instructor: Dr. Lavanya Ramapantulu, Fall-2018

Honours and Awards

- Best Paper Award FCCM'25
- EPFL EDIC Fellowship
- Dean's List for the year 2017–19, IIIT-H

Technical Skills

- Programming Languages PYTHON, C, C++, VERILOG, BLUESPEC SYSTEM VERILOG, SYSTEM VERILOG, MATLAB, BASH, OPENCL, HLS
- Tools VIVADO, VITIS HLS, XILLYBUS, AWS F1 FPGA, VTR, RUNAI

Extracurricular Activities

- Vice President of the Association of Computer Science PhDs at EPFL
- Electronics workshop coordinator at IIIT-H

Talks

- FCCM 2026 PathSteiner: Improving PathFinder with Quasi-Optimal Steiner-Tree Initialization
- FCCM 2025 Guaranteed Yet Hard to Find: Uncovering FPGA Routing Convergence Paradox

- EPFL 2024 Accelerating ASIC Emulation and Prototyping
- ICCAD 2023 IIBLAST: Speeding Up Commercial FPGA Routing by Decoupling and Mitigating the Intra-CLB Bottleneck
- HiPEAC 2023 An FPGA Overlay for CNN Inference with Fine-grained Flexible Parallelism
- FPL 2021 FPGA Accelerator for Stereo Vision using Semi-Global Matching through Dependency Relaxation